Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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Recommendation

Use prior completed assessment years to build and validate an expected-growth baseline, then apply that baseline to the latest completed year, **2024-2025**, to identify teacher, course, and section patterns that should receive priority review before the next cycle. The stakeholder metric is **BOY/EOY score gain**: end-of-year score minus beginning-of-year score for the same student record.

Historical years establish expected growth. The latest completed year is scored against that expectation to produce current review priorities.

How To Read The Model Results

The model is designed for **group-level review**, not individual student forecasting. Individual growth varies because similar students can improve by different amounts for reasons not fully captured in the extract. For that reason, the latest-year individual gain R-squared of **0.115** is a supporting model-quality measure, not the primary decision measure.

The decision question is group-level: after adjusting for starting score, readiness, attendance, prior history, course track, grade level, and section composition, did a teacher, course, or section produce more or less average growth than expected? That comparison is more stable because student-level noise partly averages out across groups.

The EOY R-squared of **0.933** is reported only to show that final score is more directly related to BOY score. It should not be read as evidence that the model precisely predicts improvement. The relevant evidence is out-of-sample lift versus a naive baseline, aggregate fit, residual calibration, uncertainty intervals, and flag stability.

The baseline selected for operations is **Growth ensemble discovery weighted**. It predicts **score gain directly** using a Validation ensemble specification. The selected tuning choice is a validation-weighted blend of boosting, ranger forest, MARS, GAM, regularized regression, and engineered-feature predictions. Against a naive prior-year mean-growth baseline, the selected model improves temporal RMSE by **8.5%** and latest-year RMSE by **6.3%**.

Under the validation framework, this model is best described as **review-priority** evidence: 4 of 7 primary gates passed. The model is useful for structured review and prioritization, while the final interpretation should remain a review process rather than a stand-alone score.

Because decisions are made at aggregate review levels, the planning-level fit is more relevant than the individual gain R-squared: latest-year gain R-squared is **0.263** for section means, **0.524** for course means, and **0.663** for teacher means.

The workflow uses the direct-growth model to create a fair expected-growth baseline, then identifies aggregate teacher, course, and section residuals with bootstrap uncertainty checks and a mixed-effects shrinkage review.

Slice	Priority or watch rows	Total reviewed
Teachers	1	5
Courses	3	9
Sections	8	24

Decision	Slice	Target	N	Gap	95% CI	q
Watch	Section	S19	12	-2.71	-5.14 to 0.01	0.320
Watch	Section	S04	9	-2.55	-6.74 to 2.41	0.505
Watch	Section	S21	12	-2.48	-4.99 to -0.18	0.267
Watch	Section	S16	10	-2.04	-4.92 to 0.41	0.400
Watch	Section	S23	9	-1.78	-4.28 to 0.68	0.434
Watch	Section	S17	9	-1.57	-5.06 to 1.99	0.505
Watch	Section	S05	11	-1.49	-4.09 to 1.53	0.505
Watch	Section	S22	10	-1.46	-3.54 to 0.91	0.505

The labels are review priorities, not personnel ratings or automated decisions.

Plain-English Method

1. Build a paired BOY/EOY growth extract from public-safe assessment records.
2. Define the business outcome as score gain: EOY score minus BOY score.
3. Use prior completed years to engineer pre-outcome predictors, compare candidate models, and select the expected-growth baseline.
4. Hold out the latest completed year as the action-year review period.
5. Score latest-year records against the prior-year baseline.
6. Aggregate observed-minus-expected growth by teacher, course, and section.
7. Assign review-priority labels when the gap is large enough to matter and the uncertainty checks support follow-up.

This design separates the prediction problem from the decision problem. The prediction model estimates what growth would be expected for a similar starting profile; the review layer asks where actual growth departed from that expectation.

Direct Answers

1. The analysis covers 1,737 paired BOY/EOY records across 174 section-year groups and 5 teacher identifiers.
2. The training window is 2018-2019, 2019-2020, 2020-2021, 2021-2022, 2022-2023, 2023-2024; the action year is 2024-2025.
3. The average raw gain across the full extract is 5.72 points; the latest-year raw gain is 5.34 points.
4. The model search tested 31 candidate baselines across parametric, nonlinear, ensemble, and excluded ID-benchmark families.
5. The selected direct-growth baseline has temporal expected-gain RMSE 4.576, temporal MAE 3.649, latest-year RMSE 4.646, and latest-year MAE 3.714.

6. The individual gain R-squared is a supporting model-quality measure; the review decision is based on group-level observed-minus-expected growth.
7. Teacher, course, and section flags are review priorities for planning and follow-up, not causal claims or personnel ratings.

Data Audit

A record enters the growth model only when the same public-safe student has valid BOY and EOY scores in the same section and teacher context. This keeps improvement tied to one instructional experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique public-safe teacher identifiers	5
Records with prior-year history	1,066
Mean section size	10.4
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

Model Selection

The model discovery system used 1,485 prior-year pairs and held out 252 latest-year pairs for action-year evaluation. The primary selection metric is stable rolling-origin temporal expected-gain RMSE, not latest-year performance, so each validation year is treated like the future. Candidate-selection folds require at least three prior completed years.

The best eligible direct-growth rolling-origin RMSE was **4.576** from **Growth ensemble discovery weighted**, so it is the operating baseline. Repeated-CV RMSE remains a stability check for candidates that are practically tied on rolling-origin RMSE.

The lowest overall temporal RMSE was **4.548** from **Growth stacked ensemble**, but that row is reported as a benchmark rather than an operating baseline because it is not an eligible direct-growth candidate.

The model-search guardrails were:

- Use direct BOY/EOY score gain as the operating target because that is the stakeholder performance metric.
- Select by stable rolling-origin temporal RMSE so the baseline is judged on future-facing generalization after enough history exists.
- Mark candidates with failed temporal folds as unstable rather than treating implausible extrapolations as credible model comparisons.
- Use repeated-CV RMSE as the tie-breaker when rolling-origin RMSE differs by less than 0.01 points.

- Keep teacher, course, and section identifiers out of the operating baseline because those are the groups being reviewed.
- Use feature engineering only when the feature is available at BOY or from prior completed years.
- Refit the selected production model on all completed years only after model selection.
- Report individual gain R-squared as a supporting model-quality measure, not as the primary decision measure.
- Report EOY R-squared only as context because final score is more directly related to starting score than growth is.

The validation targets below are conservative. They keep the interpretation aligned with the strength of the evidence.

Gate	Decision-grade	Actual	Status
Rolling RMSE lift vs naive	10.0%	8.5%	Below decision-grade
Rolling MAE lift vs naive	8.0%	10.1%	Pass
Temporal gain R-squared	0.200	0.160	Below decision-grade
Section mean gain R-squared	0.300	0.263	Below decision-grade
Course mean gain R-squared	0.500	0.524	Pass
Teacher mean gain R-squared	0.600	0.663	Pass
Overall residual bias	0.200	-0.313	Review
Maximum subgroup residual bias	0.500	3.393	Review

The first strength check is whether the selected model beats a naive baseline that predicts the training-year mean gain for every record. This is the minimum bar for using a model as an expected-growth baseline.

Measure	Value
Naive temporal RMSE	5.002
Selected temporal RMSE	4.576
Temporal RMSE improvement	8.5%
Naive latest-year RMSE	4.960
Selected latest-year RMSE	4.646
Latest-year RMSE improvement	6.3%
Naive latest-year MAE	4.026
Selected latest-year MAE	3.714
Latest-year MAE improvement	7.8%
Latest-year gain R-squared	0.115
Latest-year EOY R-squared	0.933
Section-mean gain R-squared	0.263
Teacher-mean gain R-squared	0.663
Course-mean gain R-squared	0.524

The table below shows the strongest stable candidate baselines tested. The selected model was chosen by rolling-origin validation, not by whichever model looked best on the latest year. Excluded ID benchmarks and unstable temporal candidates remain in the generated artifacts, but they are not eligible operating baselines.

Candidate model	Model type	Used?	Rolling RMSE	Latest RMSE	Latest MAE	Gain R2	Read
Discovery ensemble	Ensemble	Yes	4.576	4.646	3.714	0.115	Selected
GBM 1	Boosting		4.603	4.727	3.814	0.084	Near tie
Ensemble weighted	Ensemble		4.604	4.661	3.738	0.109	Near tie
GBM 2	Boosting		4.608	4.695	3.780	0.097	Near tie
Growth elastic net	Regularized		4.609	4.699	3.740	0.095	Near tie
Growth lasso	Regularized		4.610	4.693	3.739	0.097	Near tie
Ensemble balanced	Ensemble		4.614	4.647	3.719	0.115	Near tie
MARS 2	MARS		4.626	4.618	3.714	0.126	Near tie

Full model artifacts: model comparison, model search grid, model strength, family summary, selection rationale, temporal validation, rolling-origin validation, process validation, locked holdout, validity targets, and bootstrap validation.

Feature Discovery

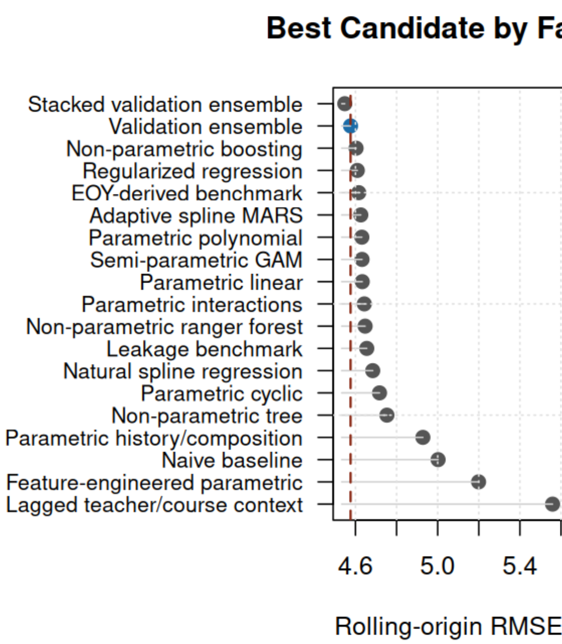
The model search includes time-appropriate feature engineering: multi-year prior student growth, prior trend and volatility, BOY score bands, section composition, attendance mix, nonlinear basis terms, and selected interactions. The table below shows which features mattered most when permuted in the locked action-year data.

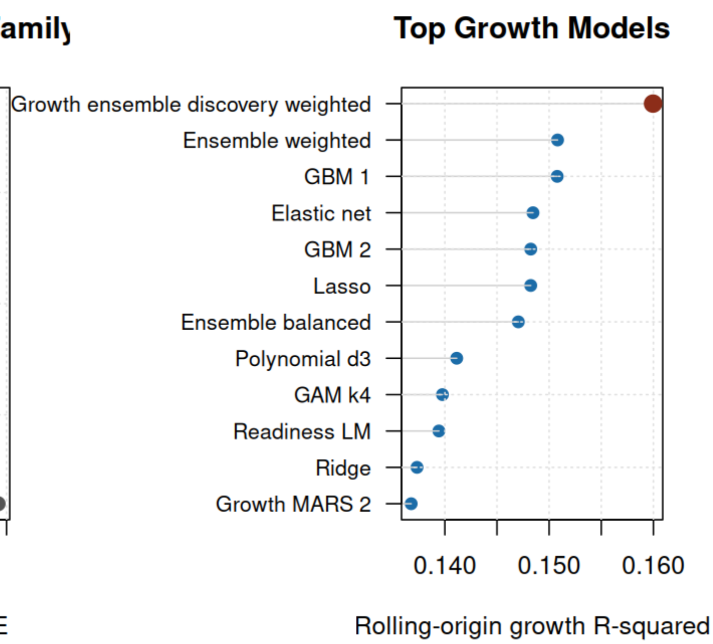
Feature	RMSE lift if permuted	SD
boy score	1.960	0.302
boy readiness	0.440	0.097
student prior mean eoy	0.263	0.064
course track	0.034	0.020
grade level	0.021	0.010
student prior year count	0.007	0.007
section student prior mean gain	0.006	0.003
section prior gain mean	0.005	0.008
section readiness mean	0.004	0.004
section boy mean	0.003	0.005

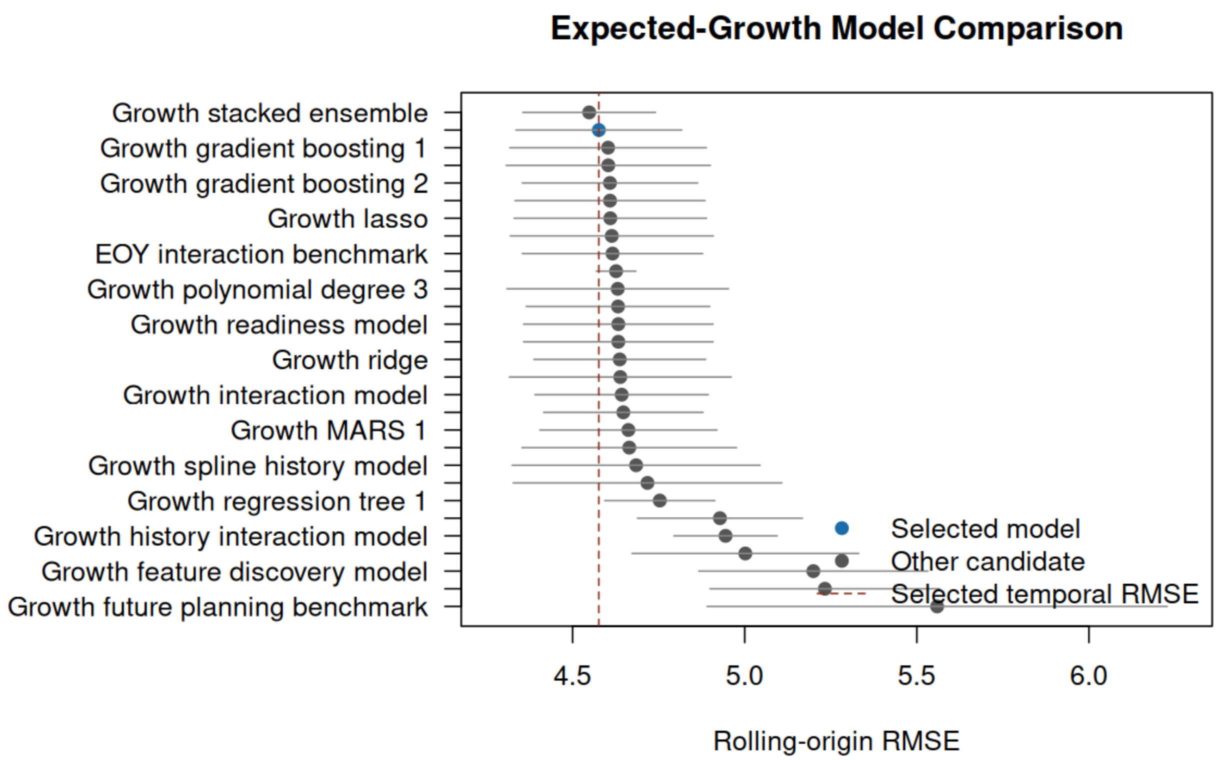
Feature stability asks whether the same predictors continue to matter across repeated perturbations.

Feature	Positive importance	Top-quartile importance
boy score	100%	100%
boy readiness	100%	100%
student prior mean eoy	100%	100%
course track	100%	88%
grade level	100%	88%
student prior year count	88%	25%
section student prior mean gain	100%	0%
section prior gain mean	62%	25%
section readiness mean	88%	12%
section boy mean	75%	12%

Feature artifacts: feature importance and feature stability.






Expected-Growth Model Comparison
Growth stacked ensemble
Growth gradient boosting 1
Growth gradient boosting 2
Growth lasso
EOY interaction benchmark
Growth polynomial degree 3
Growth readiness model
Growth ridge
Growth interaction model
Growth MARS 1
Growth spline history model
Growth regression tree 1
Growth history interaction model
Growth feature discovery model
Growth future planning benchmark
Selected model
Other candidate
Selected temporal RMSE
4.5
5.0
5.5
6.0
Rolling-origin RMSE

Latest-Year Review Targets

The latest-year review layer compares observed gain with expected gain for 24 section groups. The review labels use a practical threshold, bootstrap interval direction, BH-adjusted q-values for multiple-review control, flag stability, and a mixed-effects shrinkage check.

Review	Slice	Target	N	Gap	Stability	Shrinkage	Evidence
Priority review	Course	Precalc	40	-1.38	Directional	In range	Directional; context review
Priority review	Teacher	TCH-005	68	-1.08	Directional	In range	Directional; context review
Bright spot	Course	AP Precalc	30	+1.69	Stable	In range	Stable; shrinkage tempers
Bright spot	Section	S13	9	+2.57	Stable	In range	Stable; shrinkage tempers
Watch	Section	S19	12	-2.71	Stable	In range	Stable watch
Watch	Section	S04	9	-2.55	Stable	In range	Stable watch
Watch	Section	S21	12	-2.48	Stable	In range	Stable watch
Watch	Section	S16	10	-2.04	Stable	In range	Stable watch
Watch	Section	S23	9	-1.78	Stable	In range	Stable watch
Watch	Section	S17	9	-1.57	Directional	In range	Directional watch

Detailed review tables: teacher review, course review, and section review. Reconciled evidence: evidence reconciliation.

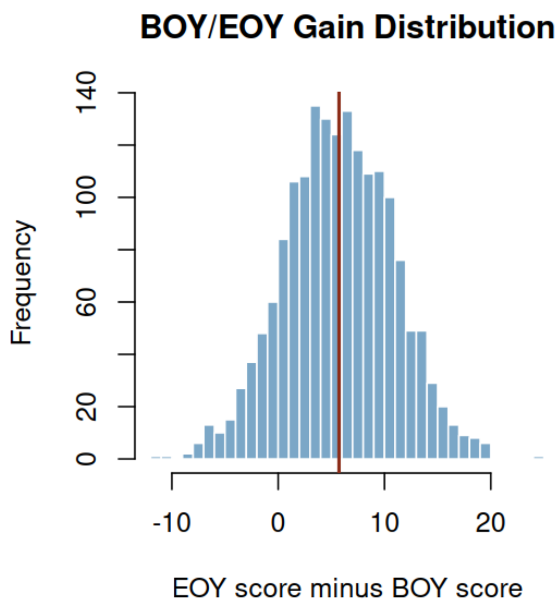
The evidence label reconciles the practical bootstrap flag with the more conservative shrinkage model. When shrinkage tempers a signal, the correct action is context review and support planning rather than escalation.

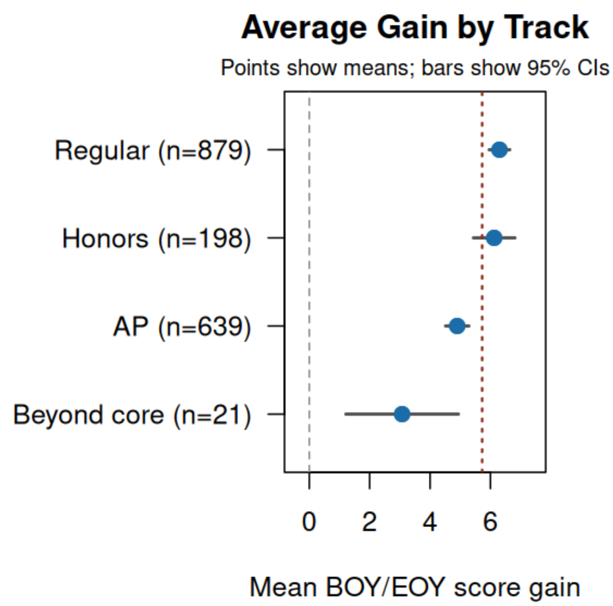
Shrinkage artifacts: shrinkage status and shrinkage review. Flag-stability artifacts: flag stability.

Section Evidence

Raw section improvement is useful for communication, but it is not the final comparison. A section can show positive raw gain and still fall below expected growth if its starting profile suggested a larger increase.

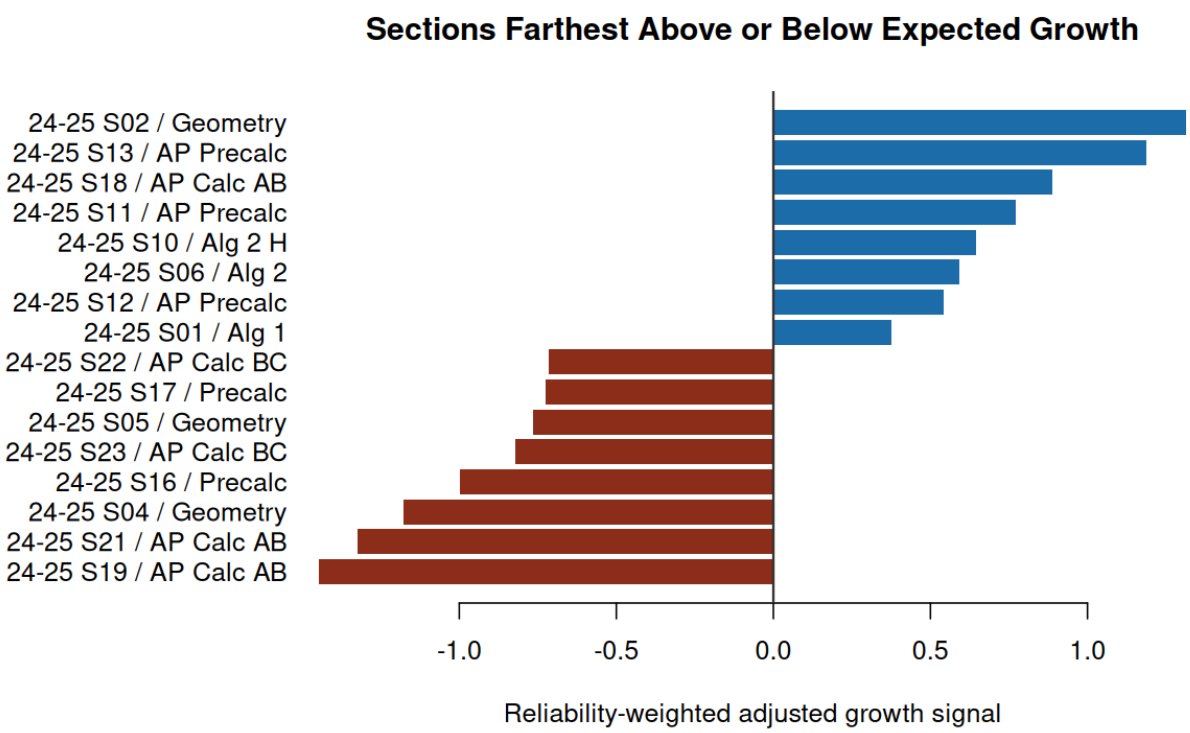
Section	Course	N	BOY	EOY	Gain	95% CI	p-value
S02	Geometry	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
S13	AP Precalc	9	63.5	70.7	7.18	5.20 to 9.16	<0.001
S18	AP Calc AB	13	49.4	56.1	6.69	3.06 to 10.31	0.002
S11	AP Precalc	10	69.0	74.7	5.67	1.52 to 9.83	0.013
S10	Alg 2 H	12	63.1	70.0	6.90	3.98 to 9.83	<0.001
S06	Alg 2	11	41.0	48.6	7.63	4.97 to 10.30	<0.001
S12	AP Precalc	11	59.5	65.1	5.64	1.86 to 9.43	0.008
S01	Alg 1	13	42.1	49.2	7.13	4.19 to 10.08	<0.001





The adjusted section signal is observed gain minus expected gain, reliability-weighted toward zero for smaller sections.

Section	Teacher	Course	N	Raw	Expected	Signal	Result
S02	TCH-002	Geometry	10	9.02	6.33	+1.31	In range
S13	TCH-004	AP Precalc	9	7.18	4.61	+1.19	Above
S18	TCH-005	AP Calc AB	13	6.69	5.08	+0.89	In range
S11	TCH-004	AP Precalc	10	5.67	4.09	+0.77	In range
S23	TCH-005	AP Calc BC	9	1.89	3.67	-0.82	In range
S05	TCH-001	Geometry	11	4.00	5.49	-0.76	In range
S17	TCH-003	Precalc	9	4.78	6.35	-0.72	In range
S22	TCH-005	AP Calc BC	10	3.10	4.56	-0.71	In range



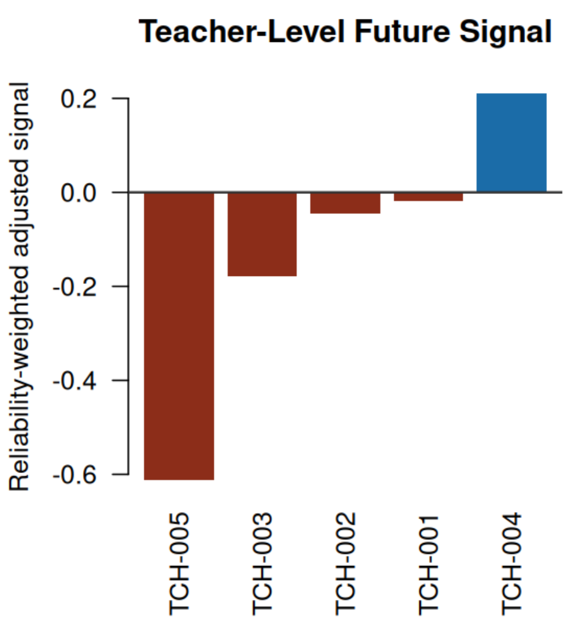
The full section signal table is generated as section signals.

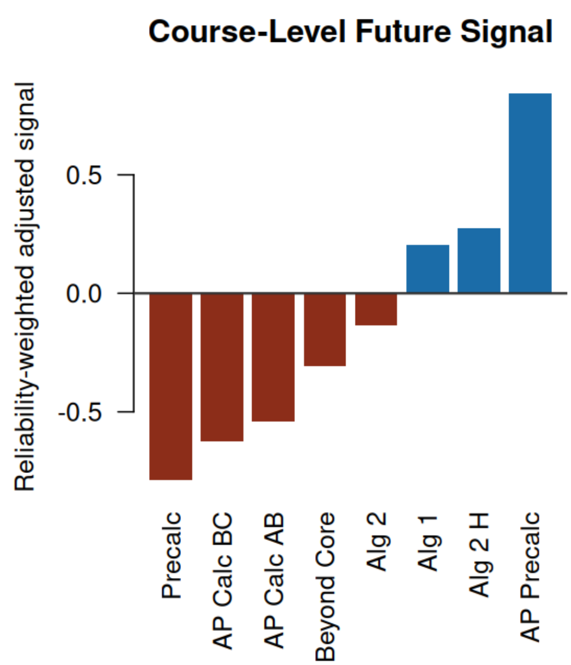
Teacher and Course Summaries

These summaries aggregate the latest-year evidence into planning views. They support review conversations about pacing, curriculum alignment, attendance mix, and transferable practices.

Teacher	Sections	Records	Raw	Expected	Signal
TCH-004	5	43	5.21	4.75	+0.21
TCH-001	3	33	6.06	6.10	-0.02
TCH-002	5	52	6.60	6.69	-0.04
TCH-003	5	56	5.70	6.05	-0.18
TCH-005	6	68	3.81	4.89	-0.61

Course	Sections	Records	Raw	Expected	Signal
AP Precalc	3	30	6.11	4.43	+0.84
Alg 2 H	2	26	6.18	5.59	+0.28
Alg 1	1	13	7.13	6.45	+0.21
Geometry	4	39	5.99	6.15	-0.09
Alg 2	3	33	6.57	6.83	-0.14
Beyond Core	1	3	-0.11	3.28	-0.31
AP Calc AB	4	49	4.31	5.18	-0.54
AP Calc BC	2	19	2.52	4.14	-0.63
Precalc	4	40	4.99	6.37	-0.79





Diagnostics and Sensitivity

The baseline is strong for final-score expectation and weaker for individual gain variation. That pattern is expected: BOY score explains much of EOY score, while individual improvement contains more unobserved classroom, attendance, motivation, and assessment variation. For review decisions, the model is used to build expected growth and then aggregate residuals by teacher, course, and section.

Diagnostic	Estimate	Interpretation
Latest-year expected-gain RMSE	4.646	Typical out-of-sample prediction error on latest-year BOY/EOY gain
Latest-year expected-gain R-squared	0.115	Share of latest-year gain variation explained by the expected-growth model
Latest-year EOY R-squared	0.933	Share of latest-year EOY score variation explained by the baseline
Latest-year residual mean	-0.313	Near 0 means expected gain is centered in the action year
Latest-year residual SD	4.644	Latest-year residual spread used for slice uncertainty

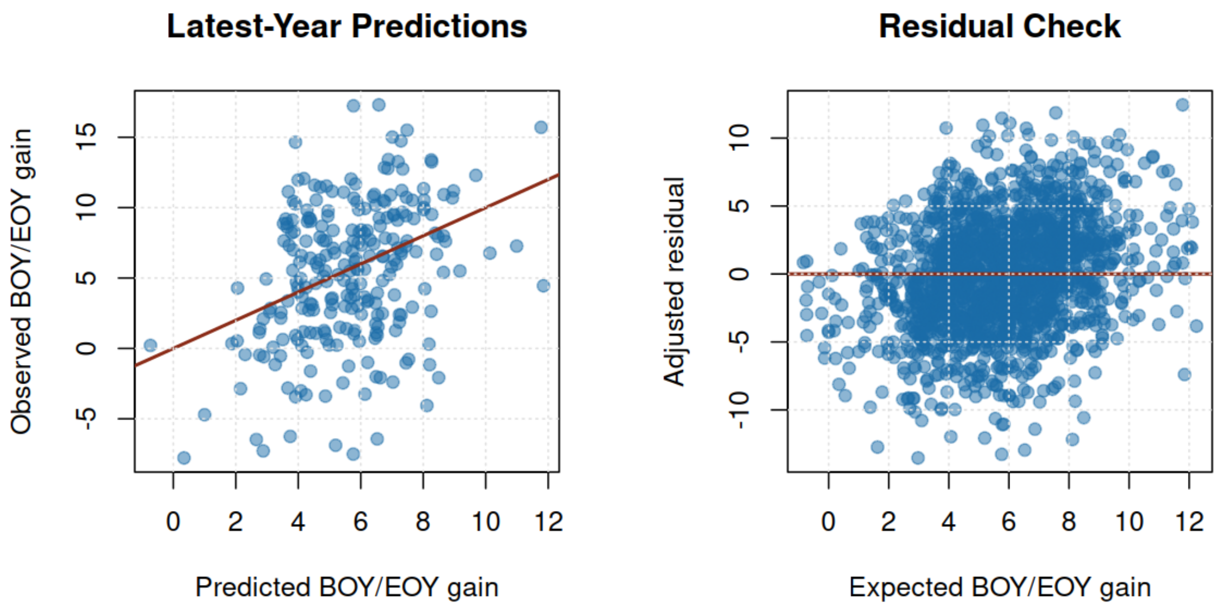
Metric	Estimate	95% interval
Expected-gain RMSE	4.646	4.254 to 5.026
Expected-gain MAE	3.714	3.355 to 4.035
Expected-gain R-squared	0.115	0.017 to 0.199
EOY RMSE	4.646	4.254 to 5.026
EOY R-squared	0.933	0.918 to 0.944

Metric	Value
Selected locked-holdout RMSE	4.646
Naive locked-holdout RMSE	4.960
Locked-holdout RMSE lift	6.3%
Selected locked-holdout MAE	3.714
Naive locked-holdout MAE	4.026
Locked-holdout MAE lift	7.8%
Locked-holdout gain R-squared	0.115

Diagnostic	Value
Individual gain SD	5.079
Student mean-gain R-squared	0.352
Section mean-gain R-squared	0.263
Course mean-gain R-squared	0.524
Teacher mean-gain R-squared	0.663

Diagnostic	Value
Selected train-validation R-squared gap	0.256

Measure	Value
Training paired records	1,485
Latest-year paired records	252
Latest-year section-year groups	24
Latest-year mean raw BOY/EOY gain	5.34
Latest-year raw-vs-adjusted rank correlation	0.865
Latest-year top-10 overlap, raw vs adjusted ranking	70.0%



Technical Appendix

The operating model excludes teacher IDs, course IDs, and section IDs. An excluded ID benchmark is reported only to show what would happen if persistent IDs were included in the baseline; it is not used for review because it would absorb the same teacher/course patterns the decision layer is designed to examine.

Package	Installed
mgcv	Available
randomForest	Available
ranger	Available
gbm	Available
rpart	Available
glmnet	Available
earth	Available

Package	Installed
lme4	Available

Metric	Value
Selected model	Growth ensemble discovery weighted
Selected target strategy	Direct growth
Selected method	Ensemble
Selected family	Validation ensemble
Selected tuned parameters	Growth gradient boosting 1 weight=2; Growth ranger forest 1 weight=2; Growth MARS 2 weight=1; Growth GAM k4 weight=1; Growth elastic net weight=1; Growth feature discovery model weight=1
Selection rule	Lowest stable rolling-origin temporal RMSE among eligible direct-growth candidates; temporal folds require at least 3 prior completed years, and repeated-CV RMSE is the tie-breaker within 0.01 points
Training paired records	1,485
Latest-year action paired records	252
Training years	2018-2019, 2019-2020, 2020-2021, 2021-2022, 2022-2023, 2023-2024
Action year	2024-2025
Candidate models tested	31
Operational candidates tested	26
Excluded ID benchmarks	5
Repeated CV folds	5
Repeated CV repeats	2
Temporal expected-gain RMSE	4.576
Temporal expected-gain MAE	3.649
Temporal expected-gain R-squared	0.160
Temporal expected-gain RMSE SD	0.241
Temporal EOY R-squared	0.931
Latest-year expected-gain RMSE	4.646
Latest-year expected-gain MAE	3.714
Latest-year expected-gain R-squared	0.115
Latest-year EOY R-squared	0.933

The full selected-model metrics table is generated as selected-model metrics.

Conclusion

The project should be read as a statistical review-priority system. The strongest business value is the workflow: choose a validated expected-growth baseline, compare latest actual growth to that baseline at the group level, quantify uncertainty by slice, reconcile competing evidence checks, and translate the evidence into review priorities.

The recommended stakeholder action is to review the flagged teacher, course, and section patterns before the next assessment cycle. Priority-review rows deserve support-oriented investigation; bright spots deserve study for transferable practices; watch-list rows deserve context review before escalation.

The important limitation is that the data are public-safe and generalized from an assessment workflow. The outputs demonstrate the analysis pattern and should be interpreted as portfolio evidence rather than operational decisions about real students or staff.

Reproducibility

Rebuild the full evidence packet with `make all`. The pipeline uses the included public-safe extract and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.